Quirk Table I reproduction: decision meeting

arXiv:1708.02243 · colored m_Q = 1.8 TeV, Λ = 2 keV · EW m_Q = 800 GeV, Λ = 4 keV · Wilson 95% intervals · p = exact binomial against the paper's value

Benchmark	Sample	k / n	ε [Wilson 95%]	Paper	p	Verdict
ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching) Same cut as the paper (Sec. III B), on the final-state quirk pair.						
Colored	COL1000	101 / 419	0.241 [0.203, 0.284]	0.24	0.95	**No statistically significant inconsistency**
	COL2000	207 / 873	0.237 [0.210, 0.266]	0.24	0.87	**No statistically significant inconsistency**
ε_{fid} = N(both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered) The paper's definition (Sec. IV); colored: both R-hadrons charged, in numerator and denominator.						
Colored	COL1000	6 / 27	0.222 [0.106, 0.408]	0.28	0.67	**No statistically significant inconsistency**
	COL2000	22 / 68	0.324 [0.224, 0.442]	0.28	0.42	**No statistically significant inconsistency**
ε_{reco} = N(plane found, 2 hits in 7 layers + ≥ 1 in the 8th, Δw < 1 cm, Δs < 0.01 cm) / N(fiducial) The paper's plane algorithm and signal-region cuts (Sec. III C, IV); SCT z not smeared.						
Colored	COL1000	5 / 6	0.833 [0.436, 0.970]	0.35	0.022	**Statistically inconsistent**
	COL2000	8 / 22	0.364 [0.197, 0.570]	0.35	1.00	**No statistically significant inconsistency**
ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching) Same cut as the paper (Sec. III B), on the final-state quirk pair.						
EW	EW1000	89 / 511	0.174 [0.144, 0.209]	0.10	2.8×10^{-7}	**Strongly inconsistent**
	EW2000	186 / 1045	0.178 [0.156, 0.202]	0.10	1.8×10^{-14}	**Strongly inconsistent**
ε_{fid} = N(both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered) The paper's definition (Sec. IV); colored: both R-hadrons charged, in numerator and denominator.						
EW	EW1000	33 / 89	0.371 [0.278, 0.475]	0.28	0.059	**Borderline / mild tension**
	EW2000	62 / 186	0.333 [0.270, 0.404]	0.28	0.12	**No statistically significant inconsistency**
ε_{reco} = N(plane found, 2 hits in 7 layers + ≥ 1 in the 8th, Δw < 1 cm, Δs < 0.01 cm) / N(fiducial) The paper's plane algorithm and signal-region cuts (Sec. III C, IV); SCT z not smeared.						
EW	EW1000	27 / 33	0.818 [0.656, 0.914]	0.72	0.25	**No statistically significant inconsistency**
	EW2000	50 / 62	0.806 [0.691, 0.886]	0.72	0.16	**No statistically significant inconsistency**

Two choices we made

- **Fiducial uses the barrel lengths of [27]** (pixel |z| < 40.05 cm, SCT |z| < 74.9 cm), the detector reference the paper names.
- **Signal uses no SCT smearing along the beam.** It is the only setting that brings both benchmarks to the paper's ε_{reco} and Δs shape. The paper's text does not state it.

Background

- Converted photon + pile-up, with the detector model the paper cites ([34]: bremsstrahlung, SCT z 580 μm): **0 / 9139** in the signal region, i.e. < 1 event at 300 fb^{-1} (95% upper limit). The paper's level is ≈ 0.7 events, so this agrees at the upper-limit level.

Finding

- Signal and background only match the paper at **different** SCT resolutions. With one realistic resolution, either ε_{reco} drops by 2.5–5× (580 μm), or the conversion background becomes ≈ 10 (our σ) to 48 (paper σ) events at 300 fb^{-1} (no smearing).

Definitions

ε_{trig} = N(pT of the quirk pair > 200 GeV) / N(events after jet matching)
ε_{fid} = N(triggered, both quirks cross all 8 barrel layers inside the [27] lengths, within 25 ns) / N(triggered)
ε_{reco} = N(fiducial, plane found, 2 hits in 7 layers + ≥ 1 in the 8th, $\Delta w < 1$ cm, $\Delta s < 0.01$ cm) / N(fiducial)
$\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

Colored: both R-hadrons must be charged; this is applied to the triggered events before ε_{fid} (27 of 101 and 68 of 207 events).

Wilson 95% interval for k of n ($\varepsilon = k/n$, $z = 1.96$): $[\ \varepsilon + z^2/2n \pm z\sqrt{(\ \varepsilon(1-\varepsilon)/n + z^2/4n^2\ )}\ ] / (1 + z^2/n)$
95% upper limit when 0 of n pass: $UL = 1 - 0.05^{1/n} \approx 3/n$
p = exact binomial probability, if the paper's value were exactly right, of a count as unlikely as ours or more

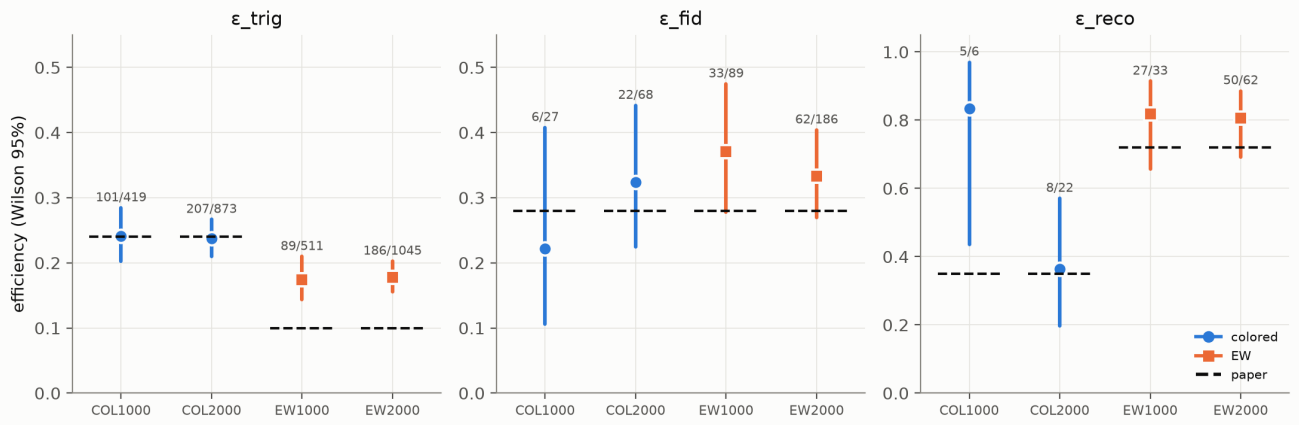
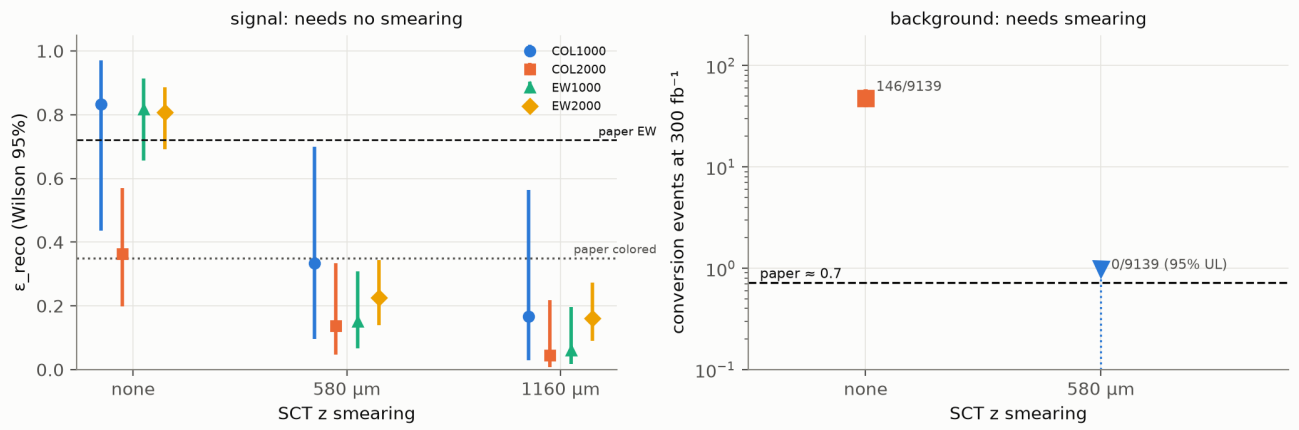
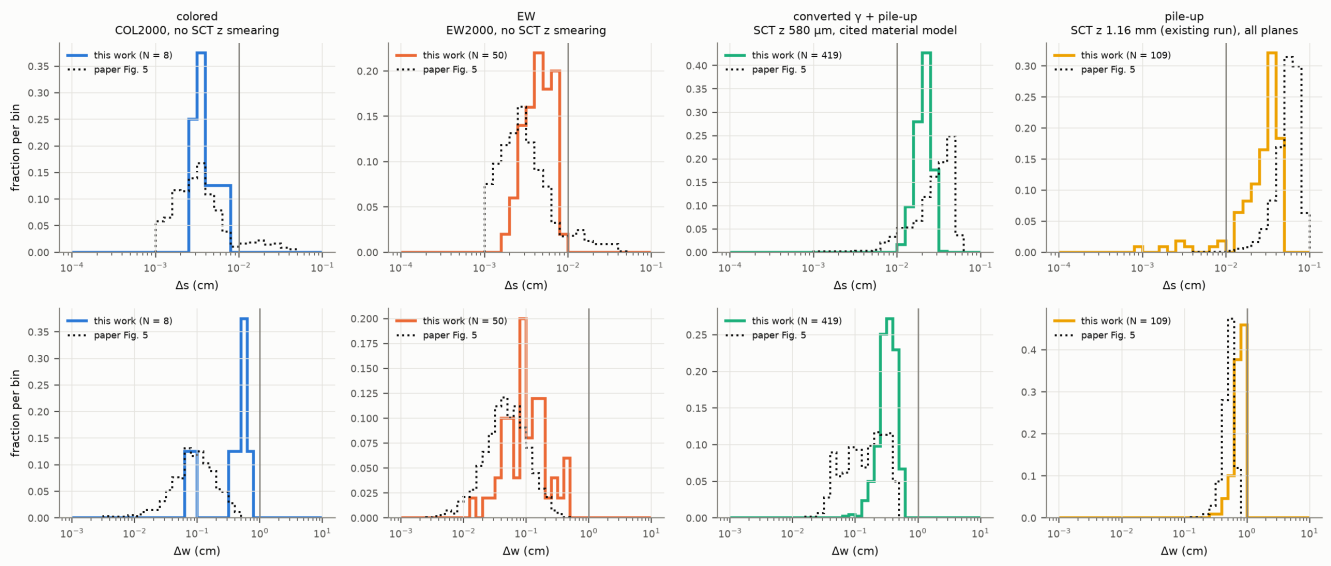


Table I efficiencies, each sample separately; bars = Wilson 95%; dashed = paper. ε_{reco} without SCT smearing along the beam.



Left: ε_{reco} for each SCT z setting. Right: converted-photon + pile-up events in the signal region at 300 fb^{-1} (paper σ ≈ 1 pb, 1% conversion); triangle = 95% upper limit for 0 events.

Paper Figure 5: Δs and Δw



Δs (top) and Δw (bottom), unit area on the paper's bins; dotted = paper's Fig. 5. Signal: no SCT smearing along the beam. Converted γ + pile-up: SCT z 580 µm with the cited material model. Pile-up: existing run (SCT z 1.16 mm), all planes. Our Δw is $\sqrt{\lambda_2}$ (paper Eq. 11); the paper's plotted Δw appears to be $\frac{1}{2}\sqrt{\lambda_2}$.

Paper Figures 2, 3 and 4

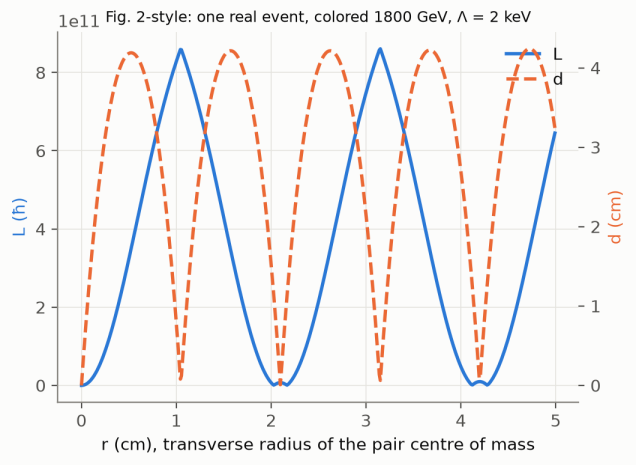


Fig. 2: angular momentum L and quirk separation d for one colored event (COL2000 event 107). L reaches ≈ 9×10^{11} ħ, the paper's ≈ 10^{12} ħ scale.

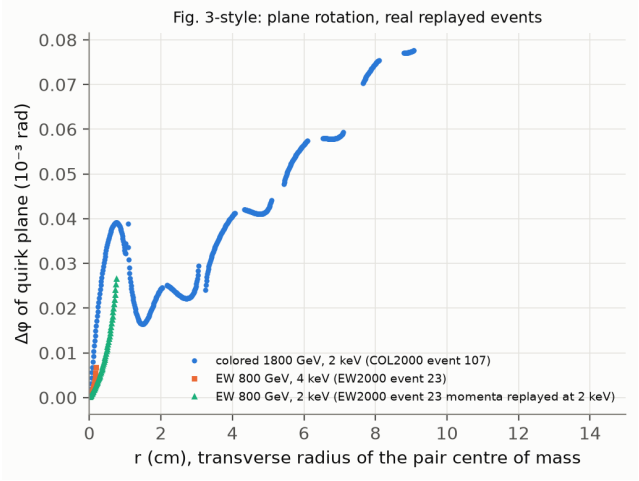


Fig. 3: rotation of the quirk plane stays below 10^{-4} rad, so the hits stay on one plane.

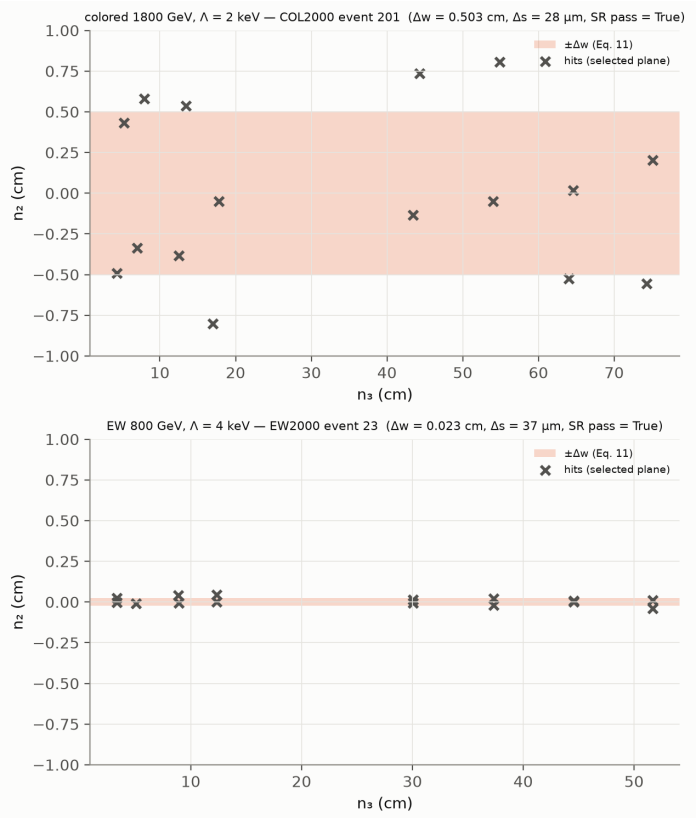


Fig. 4: hits of one colored (COL2000 event 201) and one EW (EW2000 event 23) event in the plane's own coordinates, current setup; both pass the signal region (Δs = 28 and 37 μm). Band = ±Δw.